DEEP LEARNING – HANDWRITTEN EXAM NOTES

UNIT 3: Optimization Algorithms, Adaptive Methods, AdaGrad, AdaDelta, Adam, Dropout

B.Tech Mid-1 Examination Preparation | Complete Question Bank Solutions

PART A – SHORT ANSWER QUESTIONS

Question 1

What is the main advantage of adaptive optimization methods?

Main Advantage

The main advantage is **Individual / Per-Parameter Learning Rate Adaptation**. Unlike standard SGD which applies a single global learning rate to every weight, adaptive optimizers (AdaGrad, RMSProp, Adam) dynamically scale updates based on historical gradients.

Key Benefits

- **Frequent Features:** Receive smaller updates to prevent overshooting.
- **Sparse / Infrequent Features:** Receive larger updates to ensure they are adequately learned.
- **Faster Convergence:** Drastically reduces training time and the need for tedious manual learning rate tuning.

Question 2

What is Adam?

Definition & Formula

Adam (**Ada**ptive **M**oment Estimation) is an advanced optimization algorithm that combines the best properties of **Momentum** (first moment, mean) and **RMSProp** (second moment, uncentered variance).

1st Moment (Mean): $m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$

2nd Moment (Variance): $v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2$

Bias Correction: $\hat{m}_t = m_t / (1 - \beta_1^t)$, $\hat{v}_t = v_t / (1 - \beta_2^t)$

Update Rule: $\theta_{t+1} = \theta_t - (\eta / (\sqrt{(\hat{v}_t)} + \varepsilon)) \cdot \hat{m}_t$

Question 3
What is AdaGrad?

Definition & Working

AdaGrad (**Ada**ptive **Grad**ient Algorithm) adapts the learning rate to the parameters by dividing the learning rate by the square root of the sum of all historical squared gradients.

$$G_t = G_{t-1} + g_t^2$$
$$\theta_{t+1} = \theta_t - (\eta / (\sqrt{}(G_t + \varepsilon))) \cdot g_t$$

Key Drawback: Since G_t keeps accumulating positive squared terms, the effective learning rate eventually drops to zero, prematurely stopping learning.

Question 4
What is AdaDelta?

Definition & Key Innovation

AdaDelta is an extension of AdaGrad designed to fix its aggressive, decaying learning rate problem.

- **Exponentially Decaying Average:** Instead of summing all past gradients, it restricts the window using an exponential decay ($E[g^2]_t = \rho E[g^2]_{t-1} + (1-\rho)g_t^2$).
- **No Learning Rate Hyperparameter:** AdaDelta completely eliminates the global learning rate η by replacing it with the RMS of past parameter updates $RMS[\Delta\theta]$.

Question 5
What is the purpose of dropout?

Purpose and Mechanism

Dropout is a powerful regularization technique used to **prevent overfitting** in deep neural networks.

- **Breaking Co-Adaptation:** By randomly dropping neurons (with probability $p \approx 0.5$), neurons cannot rely on specific neighboring features, forcing each neuron to learn robust, independent features.
- **Ensemble Effect:** Each training step samples a different thin architecture. At test time, using all neurons scaled by $(1-p)$ approximates the geometric mean of 2^N sub-networks.

Question 6

List out various Optimization methods

1. **Stochastic Gradient Descent (SGD)** — Fundamental first-order iterative update algorithm.
2. **SGD with Momentum** — Accelerates in the consistent direction and dampens oscillations.
3. **Nesterov Accelerated Gradient (NAG)** &mathematical "look-ahead" momentum.
4. **AdaGrad (Adaptive Gradient)** — Per-parameter learning rates based on squared gradient sum.
5. **RMSProp (Root Mean Square Propagation)** — Exponentially weighted moving average of squared gradients.
6. **AdaDelta** — Extension of AdaGrad that eliminates the learning rate hyperparameter.
7. **Adam (Adaptive Moment Estimation)** — Combines Momentum and RMSProp with bias correction.
8. **AdamW & Nadam** — Modern extensions with decoupled weight decay and Nesterov acceleration.

PART B – ESSAY / 10-MARK QUESTIONS

Essay Question 1

What is the purpose of optimization in neural networks?

1. Core Purpose

The purpose of **optimization** in neural networks is to find the set of model parameters (weights W and biases b) that **minimize the cost / loss function J(θ)** across the entire training dataset.

2. Key Objectives of Optimization

1. **Empirical Risk Minimization (ERM):** Reduce the average training loss $J(\theta) = (1/n)\sum L(f(x_i; \theta), y_i)$.
2. **Navigating Non-Convex Landscapes:** Deep network loss surfaces contain countless saddle points, ravines, and plateaus. Optimization algorithms guide weights out of these traps.
3. **Generalization Balance:** Unlike pure mathematical optimization that only cares about training data, DL optimization aims for **generalization** (low test error).
4. **Scalability:** Enables training models with millions or billions of parameters efficiently using mini-batches.

Essay Question 2

What is an optimization algorithm? What is the role of the learning rate in optimization?

1. Definition of Optimization Algorithm

An **Optimization Algorithm** is a mathematical procedure that iteratively adjusts neural network weights and biases to minimize the loss function.

2. Critical Role of Learning Rate (η)

The **learning rate (η)** is the most important hyperparameter in deep learning. It determines the **step size** taken towards the minimum along the negative gradient direction.

Learning Rate Scenario	Effect on Learning & Loss Curve	Consequence
Too Large ($\eta > 1.0$)	Overshoots the minimum; oscillates wildly	**Divergence / NaN Errors**; fails to train
Too Small ($\eta < 10^{-5}$)	Extremely tiny weight updates	**Very slow training**; gets stuck in saddle points
Optimal Learning Rate	Steady, consistent decrease in loss	**Fast and stable convergence**

Essay Question 3

Explain AdaGrad with its working

1. Mathematical Formulation

At each time step t, given gradient $g_t = \nabla_\theta J(\theta_t)$:

Squared Gradient Accumulation: $G_t = G_{t-1} + g_t \odot g_t$

Parameter Update Rule: $\theta_{t+1} = \theta_t - (\eta / (\sqrt{(G_t + \varepsilon)})) \odot g_t$

2. Step-by-Step Working

1. **Initialization:** Initialize accumulator $G_0 = 0$ and set small smoothing term $\varepsilon \approx 10^{-8}$ to avoid division by zero.
2. **For Frequent Gradients:** Gradients are large ⇒ G_t grows rapidly ⇒ Effective learning rate $\eta/\sqrt{G_t}$ becomes small, preventing overshooting.

3. **For Sparse Gradients:** Gradients are infrequent ⇒ G_t remains small ⇒ Learning rate remains large, giving rare features a bigger boost.

Essay Question 4

Compare AdaGrad and AdaDelta

Feature / Aspect	AdaGrad (Adaptive Gradient)	AdaDelta (Adaptive Delta)
Gradient Accumulation	Sums **all past squared gradients** from step 1 to t.	Uses an **exponentially decaying average** over a sliding window.
Learning Rate (η)	Requires manual initial learning rate (e.g. 0.01).	**Eliminates learning rate entirely** (uses RMS of parameter updates).
Learning Stalling Issue	**Yes** — Learning rate decays to zero, stopping training prematurely.	**No** — Learning continues throughout training without stalling.
Decay Factor (ρ)	Not applicable (unweighted sum).	Uses decay parameter $\rho \approx 0.95$.
Best Suited For	Sparse data (NLP, text classification).	Deep CNNs, RNNs, and long training runs.

Essay Question 5

Differentiate Dropout and DropConnect

Aspect	Dropout (Srivastava et al., 2014)	DropConnect (Wan et al., 2013)
What is Zeroed Out?	**Neuron Activations (Nodes)** are randomly set to 0.	**Synaptic Weights (Connections)** are randomly set to 0.
Mathematical Form	$r = m \odot a$, $y = f(Wr)$ (m is binary mask)	$y = f((M \odot W)\ x)$ (M is binary weight mask)
Sparsity Type	Node-level sparsity (entire rows/columns dropped)	Connection-level sparsity (individual matrix elements dropped)
Ensemble Space	$2^{\|V\|}$ possible subnetworks ($\|V\|$ = number of nodes)	$2^{\|E\|}$ possible subnetworks ($\|E\|$ = number of weights)
Generalization	Special case of DropConnect where all incoming weights to a node are zeroed.	Full generalization of Dropout with higher regularization capacity.

LAST-MINUTE EXAM REVISION (UNIT 3)

Quick Formula Reference

Optimizer	Key Update Formula
SGD	$\theta_{t+1} = \theta_t - \eta\ g_t$
Momentum	$v_t = \gamma v_{t-1} + \eta g_t, \quad \theta_{t+1} = \theta_t - v_t$
AdaGrad	$\theta_{t+1} = \theta_t - (\eta / \sqrt{}(G_t+\varepsilon))\ g_t$
RMSProp	$v_t = \beta v_{t-1} + (1-\beta)g_t^2, \quad \theta_{t+1} = \theta_t - (\eta / \sqrt{}(v_t+\varepsilon))\ g_t$
Adam	$\theta_{t+1} = \theta_t - (\eta / (\sqrt{}(\hat{v}_t) + \varepsilon))\ \hat{m}_t$